

Lecture on Multiplayer Games in Adversarial Search

Course: Artificial Intelligence

Instructor: Syed Hamed Raza

Overview of the Lecture:

This lecture will focus on **Multiplayer Games** in the context of **Adversarial Search**. While much of adversarial search theory and algorithms, such as **Minimax**, are focused on two-player, zero-sum games, multiplayer games introduce more complexity because there are more than two agents (players) with competing or cooperating interests. By the end of this lecture, you will understand the challenges of multiplayer games, how adversarial search algorithms extend to multiplayer settings, and some techniques to manage these complexities.

Agenda:

1. **Introduction to Multiplayer Games**
 2. **Challenges in Multiplayer Games**
 3. **Types of Multiplayer Games**
 - **Competitive Multiplayer Games**
 - **Cooperative Multiplayer Games**
 4. **Extending Minimax to Multiplayer Games**
 5. **Algorithms for Multiplayer Games**
 6. **Classroom Discussion and Problem Solving**
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1. Introduction to Multiplayer Games

What are Multiplayer Games?

- **Multiplayer games** involve more than two players, each having their own individual goals and strategies. Unlike two-player games where there are only two competing agents (Maximizer vs. Minimizer), multiplayer games include many agents who may either cooperate, compete, or exhibit both behaviors.
- **Example:** Games like **Monopoly**, **Risk**, and **Fortnite** are multiplayer games where several agents compete for dominance, but they can also include temporary alliances.

Key Characteristics of Multiplayer Games:

- **Multiple Players:** There are more than two agents involved.
 - **Mixed Interests:** Agents may have conflicting goals (e.g., all players trying to win), but they might also have cooperative elements (e.g., temporary alliances).
 - **No Zero-Sum:** In multiplayer games, the situation is rarely zero-sum. In other words, one player's gain does not necessarily result in another player's loss.
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2. Challenges in Multiplayer Games

Increased Complexity:

- The primary challenge with multiplayer games is that the number of possible strategies grows exponentially as more players are added. In a two-player game, the decision-making process is somewhat straightforward—one player tries to maximize their utility while the other tries to minimize it.
- In multiplayer games, the decision-making must consider all other players' actions and interactions, increasing the search space considerably.

Strategy Complexity:

- Players must not only evaluate their own strategies but also anticipate the strategies of multiple other players.
- This requires more advanced algorithms, as predicting outcomes becomes far more complex. There may also be non-obvious alliances, shared resources, or multiple phases in the game that affect the choices each player makes.

Non-Deterministic Outcomes:

- Some multiplayer games introduce **non-determinism**, where a player's outcome depends on both their own actions and the actions of others. This uncertainty makes it harder to plan optimal strategies.
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3. Types of Multiplayer Games

Competitive Multiplayer Games

- In **competitive multiplayer games**, all players are essentially adversaries, each aiming to maximize their own utility at the expense of others. There is no cooperation or temporary alliances, and the goal is to "win" over all other players.
- **Examples:**
 - **Risk:** A board game where players try to conquer territories and eliminate opponents.
 - **Battle Royale games** (e.g., **Fortnite**, **PUBG**): Players compete individually against many others until only one remains.
- **Adversarial Search in Competitive Games:** These games are often treated as generalized **adversarial search** problems, where every player is treated as an opponent. Strategies are based on the idea that every other player is trying to reduce the player's chances of winning.

Cooperative Multiplayer Games

- In **cooperative multiplayer games**, players work together to achieve a shared goal, often against the environment or other forces in the game. There may still be competition for rewards, but the primary focus is on cooperation.
- **Examples:**

- **Pandemic:** Players cooperate to stop the spread of diseases.
 - **Left 4 Dead:** A cooperative survival game where players work together to fight against zombies.
 - **Adversarial Search in Cooperative Games:** In cooperative multiplayer settings, adversarial search algorithms are less frequently applied since players aim to maximize their collective outcome, not individual outcomes. However, some games may still have elements of competition or mixed goals that require search algorithms to balance cooperation and competition.
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4. Extending Minimax to Multiplayer Games

While the **Minimax Algorithm** is designed for two-player games, it can be extended to multiplayer games by adapting the concept of optimal play. Instead of simply alternating between a MAX and MIN player, the algorithm must now consider multiple players and possible outcomes.

N-Player Minimax

- In **N-player Minimax**, instead of minimizing and maximizing between two players, the algorithm alternates between the **players** in the game, each choosing an optimal strategy based on their own goal.
- The challenge in extending the Minimax algorithm to N-players is the **increased branching factor** of the game tree.

Minimax with Multiple Agents

- Each agent must consider the actions of all other players, and the tree grows quickly. For example, if each player has a set of possible moves at each node, the total number of possible game states grows exponentially.
 - Players must make decisions not only based on the current state of the game but also based on predicting the moves of other players.
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5. Algorithms for Multiplayer Games

Maxn Algorithm

- **Maxn** is an extension of the Minimax algorithm that is specifically designed for multiplayer games. Instead of just alternating between two players (MAX and MIN), **Maxn** allows for multiple players, each with their own utility function.
- In **Maxn**, each player chooses the move that maximizes their own outcome, considering the strategies of all other players. The algorithm works by recursively exploring the game tree and evaluating the utility for each player at each decision point.

Monte Carlo Tree Search (MCTS)

- **MCTS** is a non-deterministic search algorithm that is well-suited for multiplayer games. It uses random simulations to estimate the potential outcomes of different moves, rather than relying on a complete evaluation of the game tree.

- MCTS can be used in multiplayer settings, especially in games with large or continuous state spaces where exhaustive search methods (like Minimax) are impractical.
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6. Classroom Discussion and Problem Solving

- **Open Discussion:** Discuss how multiplayer games such as **Risk** and **Fortnite** are modeled using adversarial search techniques. What are the unique challenges when applying algorithms like **Maxn** to multiplayer games?
 - **Hands-on Activity:** Implement **Maxn** for a simple multiplayer game (e.g., a modified version of Tic-Tac-Toe or a game with three players) and explore the challenges involved in managing multiple players.
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Conclusion:

- **Summary:** In this lecture, we explored the complexities of **multiplayer games** in the context of adversarial search. We discussed the challenges of increasing complexity as more players are added, the distinction between competitive and cooperative multiplayer games, and how search algorithms like **Maxn** and **Monte Carlo Tree Search** are used to handle these challenges.
- **Next Steps:** In the next lecture, we will dive deeper into algorithms like **MCTS** and explore their application to real-time multiplayer games and environments with imperfect information.